

Phase 2 SQL Data Requirement Document

Official Website and Customer Enquiry Management System for a UK Auto Parts SME

Field	Details
Client Name	Auto Parts SME
Document Owner	Business Analyst
Working Date	June 20XX
Sprint	Sprint 4 - Future SQL and Power BI Enhancement
Version	v1.0
Confidentiality Note	Anonymised portfolio sample. No real client name, URL, logo, address, phone number, email, or identifiable detail is used.

1. Purpose

This document defines the Phase 2 Structured Query Language (SQL) data requirements for the future Customer Enquiry and Sales Lead Tracking Dashboard. The purpose is to identify what enquiry data should be captured, stored, validated, and made ready for Power BI reporting in a later enhancement phase.

This is a BA requirement document, not a final database build document. The final physical database design should be confirmed by the technical team.

2. Phase 2 Scope

Phase 2 is focused on creating a simple enquiry and sales lead data structure so the business can track customer enquiries, quote requests, product category interest, lead status, follow-up notes, and enquiry source.

- Capture customer enquiry records from Request a Quote, Contact Us, Feedback, and Newsletter forms.
- Store product category selected by customers during quote enquiry.
- Track lead status such as New, Contacted, Quoted, Converted, Lost, or Closed.
- Record follow-up notes and next follow-up date.
- Prepare data fields required for Power BI dashboard measures and filters.
- Support data validation for missing contact details, duplicate leads, lead status accuracy, and dashboard number checks.

3. Out of Scope

- No online payment, checkout, customer login, live inventory, or customer dashboard.
- No real-time integration with supplier systems or third-party applications.
- No production database build in this document.
- No Power BI dashboard design in this document. Power BI requirements will be documented separately.
- No personal client identifiable details are included.

4. Data Sources

Source	Purpose	Key Data Captured	Phase 2 Use
Request a Quote Form	Customer asks for part quotation	Name, email, phone, category, vehicle details, message, consent	Main sales lead source
Contact Us Form	General customer message	Name, email, phone, message, consent	General enquiry tracking
Feedback Form	Customer sends feedback	Feedback type, rating, message, optional name/email	Service improvement insight
Newsletter Form	Customer subscribes to updates	Email, consent, subscription date	Marketing consent record

5. Proposed Data Entities

Entity	Description	Main Identifier	Created From	Business Owner
Customer	Stores customer contact information used for enquiry follow-up.	Customer_ID	Quote, Contact, Feedback forms	Business Owner
Enquiry	Stores each customer enquiry received through website forms.	Enquiry_ID	Quote, Contact, Feedback forms	Sales/Customer Service
Quote Request	Stores specific quote details where customer requests auto part pricing/availability follow-up.	Quote_Request_ID	Request a Quote form	Sales Team
Product Category	Stores agreed auto parts categories used for enquiry classification.	Category_ID	Product category list	Business Owner
Lead Status	Stores current status of each enquiry or quote request.	Lead_Status_ID	Sales follow-up process	Sales Team
Follow-up Note	Stores follow-up comments, next action, and follow-up date.	Follow_Up_ID	Manual sales update	Sales Team
Enquiry Source	Stores the source/channel from where enquiry came.	Source_ID	Website forms/referral/phone/email	Business Owner
Newsletter Subscription	Stores subscription consent and unsubscribe status.	Subscription_ID	Newsletter form	Marketing/Business Owner

6. Data Field Requirements

The table below defines the key SQL data fields required from a BA perspective. Final data types and indexing should be confirmed by the database developer.

Entity	Field Name	Business Meaning	Suggested Type	Required?	Rule/Note
Customer	Customer_ID	Unique customer identifier	Integer/UUID	Yes	System generated
Customer	Full_Name	Customer full name	Text	Yes for quote/contact	Captured from form
Customer	Email_Address	Customer email address	Text	Yes for quote/contact/newsletter	Must follow email format
Customer	Phone_Number	Customer phone number	Text	No	Optional but useful for follow-up
Enquiry	Enquiry_ID	Unique enquiry identifier	Integer/UUID	Yes	System generated
Enquiry	Enquiry_Type	Quote, Contact, Feedback, Newsletter	Text/Lookup	Yes	Used for dashboard filtering
Enquiry	Enquiry_Date	Date enquiry received	Date/Time	Yes	Used for monthly trend
Enquiry	Message_Text	Customer message/details	Text	Conditional	Required for quote/contact/feedback
Quote Request	Quote_Request_ID	Unique quote request identifier	Integer/UUID	Yes	System generated
Quote Request	Product_Category_ID	Selected auto parts category	Integer/Lookup	Yes	Links to product category
Quote Request	Vehicle_Make	Vehicle make	Text	No	Optional field
Quote Request	Vehicle_Model	Vehicle model	Text	No	Optional field

Quote Request	Vehicle_Year	Vehicle year	Number/Text	No	Optional field
Quote Request	Part_Details	Customer part requirement	Text	Yes	Required for quote
Lead Status	Lead_Status	New, Contacted, Quoted, Converted, Lost, Closed	Lookup	Yes	Default New
Follow-up Note	Follow_Up_Note	Sales follow-up note	Text	No	Added manually
Follow-up Note	Next_Follow_Up_Date	Next follow-up date	Date	No	Used for pending follow-up report
Enquiry Source	Source_Name	Website, phone, email, referral, social media	Lookup	Yes	Default Website for website forms
Newsletter	Consent_Given	Newsletter consent flag	Boolean	Yes	Must be true for subscription
Newsletter	Unsubscribe_Status	Subscribed or unsubscribed	Boolean/Text	Yes	Used for marketing consent control

7. Basic Entity Relationship Requirement

At BA level, the system should allow enquiry records to connect customer details, quote request details, product category, lead status, source, and follow-up notes.

- One Customer can have many Enquiries.
- One Enquiry can be linked to one Quote Request when the enquiry type is quote.
- One Quote Request must be linked to one Product Category.
- One Enquiry should have one current Lead Status.
- One Enquiry can have many Follow-up Notes.
- One Enquiry should have one Enquiry Source.
- One Customer can have zero or one active Newsletter Subscription record.

Shiv, you have to make this diagram in Draw.io or Visio. I will help you. This should be a simple logical data relationship diagram, not a complex developer-level database diagram.

8. Data Validation Requirements

Validation Area	Requirement	Example Check	Owner
Missing contact details	Identify enquiries missing email and phone.	Email_Address is blank and Phone_Number is blank	BA/QA/Data Analyst
Duplicate leads	Identify possible duplicate customers or repeated enquiries.	Same email and same category within short period	BA/Data Analyst
Lead status accuracy	Lead status must use approved values only.	Not in New, Contacted, Quoted, Converted, Lost, Closed	Sales Team/BA
Dashboard number validation	Dashboard totals must match source enquiry table.	Total enquiries in Power BI equals SQL count	BA/Data Analyst
Consent validation	Newsletter records require positive consent.	Consent_Given is true before subscription active	Business Owner/BA
Category validation	Quote requests must have valid product category.	Category_ID exists in Product Category table	BA/QA

9. Business Rules for SQL Data

- Every enquiry record must have a unique Enquiry_ID.
- Every quote request must have customer contact details and product category.
- Default lead status for a new enquiry should be New.
- Lead status should be updated manually by the sales or business team during follow-up.
- Newsletter subscription must not be active unless consent is captured.

- Product category values should match the agreed website categories: Body Parts, Lighting, Engines, Differential, Electrical, Gearbox, Other.
- Feedback records should not be shown publicly unless separately approved by the business.
- Personal data use and retention should be confirmed by the business owner or relevant advisor.

10. Reporting Readiness for Power BI

These SQL requirements support future Power BI dashboard measures. The Power BI dashboard requirement document will define visuals separately, but the SQL data should support the following reporting needs:

- Total enquiries received.
- Total quote requests received.
- Top product categories by enquiry count.
- Pending follow-ups.
- Lead conversion rate.
- Monthly enquiry trend.
- Enquiry source breakdown.
- Lead status summary.

11. Assumptions and Open Questions

ID	Assumption / Question	Owner
A1	Phase 2 will use sample or anonymised data for portfolio demonstration.	Business Analyst
A2	Final database platform and hosting decision will be made by technical owner.	Technical Owner
A3	Data retention and privacy handling will be confirmed by business/compliance owner.	Business Owner
Q1	Will the business manually update lead status, or will a simple internal form be added later?	Business Owner
Q2	Should duplicate lead matching use email only, phone only, or both?	Product Owner
Q3	Which enquiry sources should be mandatory for dashboard filtering?	Product Owner

12. BA Summary

The Business Analyst has defined the initial SQL data requirements for Phase 2 so that the future reporting enhancement has a clear data foundation. This document connects website enquiry forms with future lead tracking and reporting needs, without adding Phase 2 features into the Phase 1 website MVP.